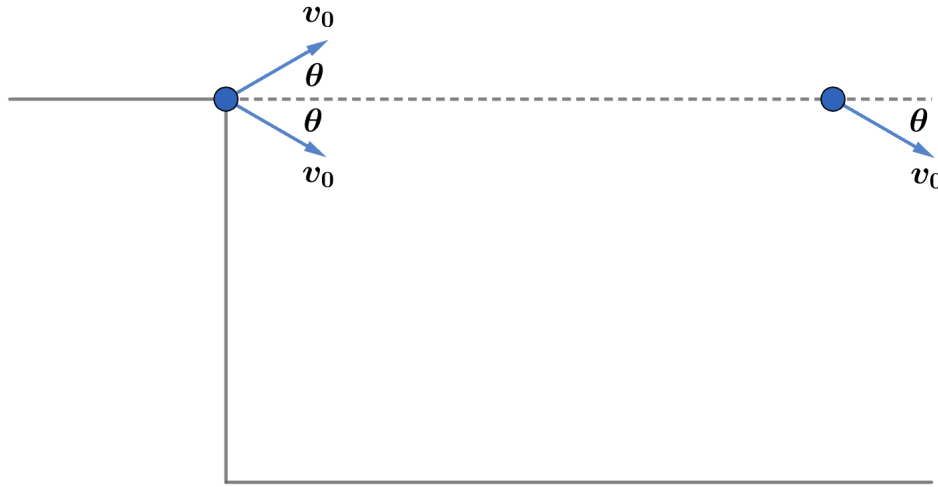


2010 F=ma Exam: Problem 5

Kevin S. Huang

Recall from projectile motion that the y-component of a particle's velocity flips direction when the particle returns to its starting height.



Applied to our case, the particle thrown upwards returns to the height of the ledge with the same velocity as the particle thrown downwards. Thus, the difference in time between the two trajectories only comes from the time the particle thrown up takes to come back down. From kinematics, the time taken to reach the peak height is

$$v_y - gt = 0$$
$$t = \frac{v_y}{g} = \frac{v_0 \sin \theta}{g}$$

Coming back down takes the same time so the total time is

$$T = 2t = \frac{2v_0 \sin \theta}{g} = \frac{2(50 \text{ m/s}) \sin(37^\circ)}{10 \text{ m/s}^2} = 6 \text{ s}$$

so the answer is C.